2021

AUSTRALIAN MATHEMATICS COMPETITION

AMC.

AUSTRALIAN MATHS TRUST

Upper Primary Grade 5-6

TIME ALLOWED

60 minutes

Instructions and Information

General

1. Do not open the booklet until told to do so by your teacher.
2. You may use any teaching aids normally available in your classroom, such as MAB blocks, counters, currency, calculators, play money etc. You are allowed to work on scrap paper and teachers may explain the meaning of words in the paper. Mobile phones are not permitted.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 000 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a competition not a test; do not expect to answer all questions. You are only competing against your own year in your own country/Australian state so different years doing the same paper are not compared.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are entered. It is your responsibility to correctly code your answer sheet.
7. When your teacher gives the signal, begin working on the problems.

The answer sheet

1. Use only lead pencil.
2. Record your answers on the reverse of the answer sheet (not on the question paper) by FULLY colouring the circle matching your answer.
3. Your answer sheet will be scanned. The optical scanner will attempt to read all markings even if they are in the wrong places, so please be careful not to doodle or write anything extra on the answer sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

Integrity of the competition

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

Reminder

You may sit this competition once, in one division only, or risk no score.

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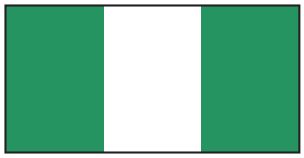
Upper Primary Division

1-10 题，每题 3 分
Questions 1 to 10, 3 marks each

1. 右图中，尼日利亚国旗上涂有白色和绿色。
请问绿色部分占总面积的几分之几？



(A) 三分之一 (B) 四分之一 (C) 二分之一 (D) 五分之二 (E) 三分之二

This Nigerian flag is white and green.
What fraction of it is green?

(A) one-third (B) one-quarter (C) one-half (D) two-fifths (E) two-thirds

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2. 请问空格内要填入哪个数才能使下面的等式成立？

$$\square - 5 = 9$$

Which number makes this number sentence true?

$$\square - 5 = 9$$

(A) 0 (B) 4 (C) 12 (D) 9 (E) 14

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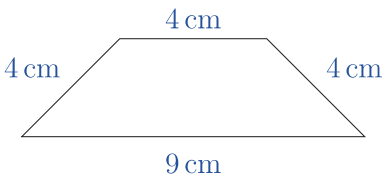
3. 请问右图中四边形的周长是多少？

What is the perimeter of the quadrilateral shown?



(A) 13 cm (B) 15 cm (C) 17 cm (D) 19 cm (E) 21 cm

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4. 请问下列选项中哪个小数最小？

Which of the following decimal numbers has the smallest value?

(A) 0.0002 (B) 0.002 (C) 0.02 (D) 0.2 (E) 2.0

5. $\frac{1}{2} + \frac{2}{4} - \frac{4}{8} =$

(A) $\frac{1}{2}$ (B) 1 (C) $1\frac{1}{2}$ (D) 2 (E) 4

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6. 小苏有很多 20 美分和 50 美分的硬币。请问她**无法**凑出以下哪个金额？

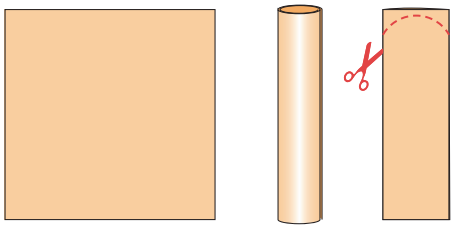
(A) 50 美分 (B) 60 美分 (C) 80 美分 (D) 30 美分 (E) 70 美分

Suri has a number of 20-cent and 50-cent coins. Which of the following amounts of money is it **NOT** possible for her to make?

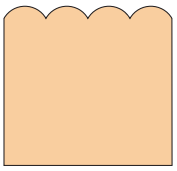
(A) 50 cents (B) 60 cents (C) 80 cents (D) 30 cents (E) 70 cents

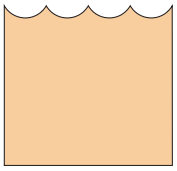
---

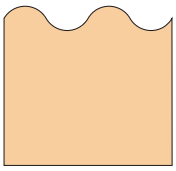
7. 将一张正方形纸张卷起来后压平，然后按照如图所示的方式剪开。
把它展开平放在桌面，请问这一张纸看起来像以下哪种形状？

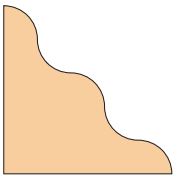


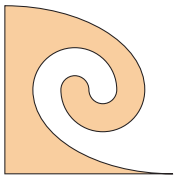
A square of paper is rolled up, pressed flat, and then cut as shown.
What could the sheet of paper look like when unrolled and laid flat?

(A) 

(B) 

(C) 

(D) 

(E) 

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8. 小利正在学校排队。有四名学生排在他前面，他后面的学生是前面的两倍。请问这列队伍一共有多少名学生？

Leo is waiting in line at school. There are four students ahead of him and twice as many behind him. How many students are in this line?

(A) 4 (B) 8 (C) 9 (D) 12 (E) 13

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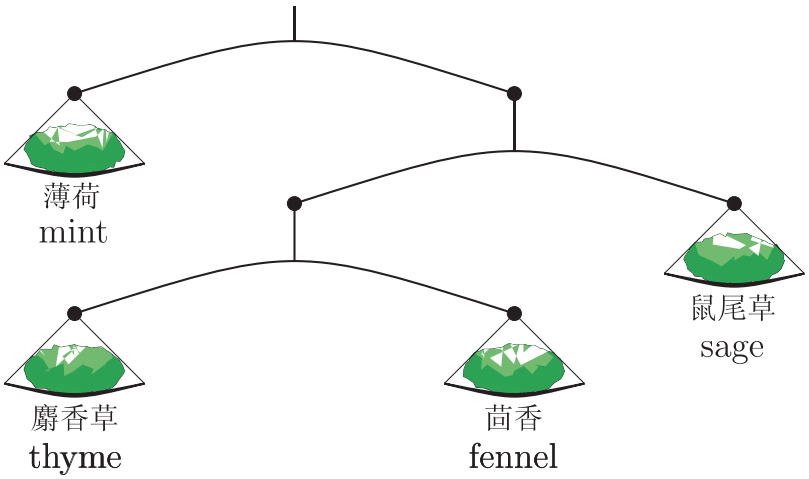
9. 小恺混合了一些草药来制作医疗药水。她利用下面这座天平来称草药。如果她使用 5 克的茴香，请问她需要多少克薄荷？



Cassandra makes a healing potion from a mixture of herbs. She uses this balance to weigh out the herbs. If she uses 5 grams of fennel, how many grams of mint will she need?

(A) 5 (B) 10 (C) 15 (D) 20 (E) 40

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10. 一个果盆中共有 14 个水果。已知桃子的数量是梨子的两倍，桃子的数量是苹果的一半，且没有其它种类的水果。请问果盆中有多少个苹果？

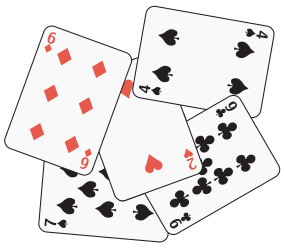
There are 14 pieces of fruit in a bowl. There are twice as many nectarines as pears, and half as many nectarines as apples. There are no other types of fruit. How many apples are there?

(A) 2 (B) 4 (C) 6 (D) 8 (E) 10

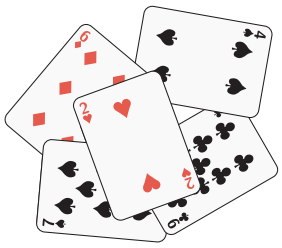
11-20 题，每题 4 分
Questions 11 to 20, 4 marks each

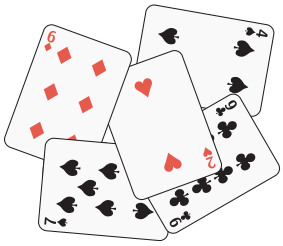
11. 我正在洗一副扑克牌，但时不时会掉一张牌在地上。过了一阵子，我才注意到已经掉了五张牌。
从上方看去，这五张牌的位置可以是以下选项中的一张图，请问可能是哪一张图？

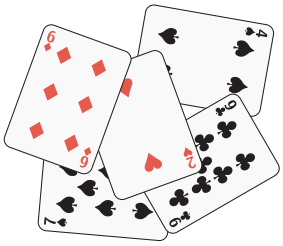
I am shuffling a deck of cards but I accidentally drop a card on the ground every now and then. After a while, I notice that I have dropped five cards.
From above, the five cards look like one of the following pictures. Which picture could it be?

(A) 

(B)

(C) 

(D) 

(E) 

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12. 右图这个矩形由两个正方形连接而成。
已知每个正方形的面积为 $25\,\text{cm}^2$。
请问这个矩形的周长是多少？

This rectangle has been made by joining two squares together.
Each square has an area of $25\,\text{cm}^2$.
What is the perimeter of the rectangle?

(A) 18 cm (B) 20 cm (C) 26 cm (D) 30 cm (E) 50 cm

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13. 一只大袋鼠正在追逐位于它前方 42 m 处的一只小袋鼠。这只大袋鼠每次向前跳 4 m，小袋鼠每次向前跳 1 m。请问这只大袋鼠要跳多少次才能追上小袋鼠？

A kangaroo is chasing a wallaby that is 42 metres ahead. For every 4-metre hop the kangaroo makes, the wallaby makes a 1-metre hop. How many hops will the kangaroo have to make to catch up with the wallaby?

(A) 8 (B) 10 (C) 11 (D) 14 (E) 21

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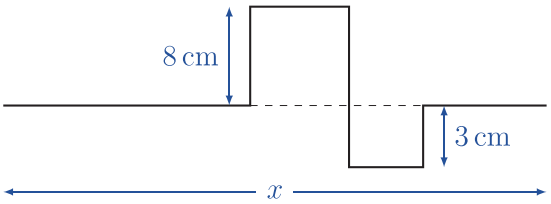
14. 一段直铁丝的总长度为 50cm。在这段铁丝上折出六个直角，使得最终的形状看起来如下图所示：



其中两小段的长度已经在图中标出。请问标注 x 的长度是多少？

A piece of straight wire is 50 cm long. Six right-angled bends are made in the wire, so that it ends up looking like the diagram shown:

The lengths of two sections are shown. What is the length marked x?

(A) 28 cm (B) 31 cm (C) 34 cm (D) 36 cm (E) 39 cm

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15. 小玛和小萝都住在火车站附近。两人都打算搭乘上午 10 点的火车。小玛认为她的手表快了 10 分钟，但实际上它是慢了 10 分钟。小萝认为她的手表慢了 10 分钟，但实际上它是快了 5 分钟。如果两人离开家去赶火车都不想在站台上等待，请问谁会错过火车且迟到了多长时间？

(A) 小玛迟到了 10 分钟　(B) 小玛迟到了 20 分钟
(C) 小萝迟到了 5 分钟　(D) 小萝迟到了 15 分钟
(E) 两人都没有迟到

Margie and Rosie both live near Lawson train station. Each plans to catch the 10 am train. Margie thinks her watch is 10 minutes fast, but in fact it is 10 minutes slow. Rosie thinks her watch is 10 minutes slow, but in fact it is 5 minutes fast. Each of them leaves home to catch the train without having to wait on the platform. Who misses the train, and by how much?

(A) Margie by 10 minutes　(B) Margie by 20 minutes
(C) Rosie by 5 minutes　(D) Rosie by 15 minutes
(E) Neither of them

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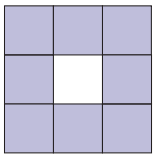
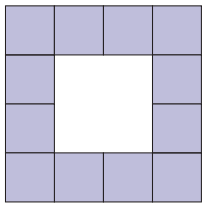
16. 小莎正在搭积木并搭出了一个她称之为*空心正方形* 的形状。这些正方形的形状都相同。

空心正方形 1



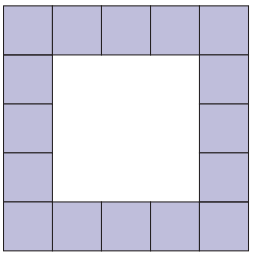
Hollow Square 1

空心正方形 2



Hollow Square 2

空心正方形 3



Hollow Square 3

请问她需要多少个方块来构造空心正方形 7？

Sally was playing with block patterns and came up with this one she called *Hollow Squares*. They all follow the same pattern.

How many blocks would she need to make Hollow Square 7?

(A) 28　(B) 30　(C) 32　(D) 34　(E) 53

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17. 我的水壶中装有 100 mL 液体，其中一半是醋，另一半是橄榄油。请问需要再添加多少醋才能使得壶中混合液体有三分之一是橄榄油？

I have a jug containing 100 mL of liquid, which is half vinegar and half olive oil. How much vinegar must I add to make a mixture which is one-third olive oil?

(A) 30 mL　(B) 40 mL　(C) 50 mL　(D) 60 mL　(E) 100 mL

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18. 现在是上午 10 点，请问经过 2021 个小时后是几点？

(A) 上午 11 点　(B) 下午 1 点　(C) 下午 3 点　(D) 下午 4 点　(E) 下午 5 点

It is 10 am now. What time will it be in 2021 hours time?

(A) 11 am　(B) 1 pm　(C) 3 pm　(D) 4 pm　(E) 5 pm

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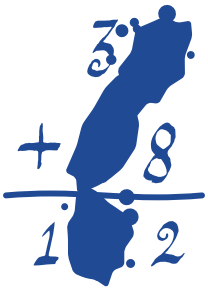
19. 小亚的钢笔水漏在了他的加法作业上，覆盖了算式中的三个数字，如图所示。请问有多少种不同的可能可以使得此算式成立？

Alexander's pen leaked on his addition homework, covering up three of the digits in the calculation shown. How many different possibilities are there for the correct working?



(A) 2　(B) 3　(C) 4　(D) 5　(E) 6

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20. 学校正在筹办一场益智问答之夜，预计会有 25 至 35 人前来参加，这些人会被分为多个队伍，每支队伍 6 至 8 人。
请问队伍数的可能范围是什么？

(A) 4 至 5　(B) 4 至 6　(C) 5 至 6　(D) 3 至 6　(E) 3 至 5

Our school is organising a quiz night. They are expecting from 25 to 35 people to come. The people will be arranged in teams of 6 to 8 people.
What is the range of possible numbers of teams to expect?

(A) 4 to 5　(B) 4 to 6　(C) 5 to 6　(D) 3 to 6　(E) 3 to 5

21-25 题，每题 5 分
Questions 21 to 25, 5 marks each

21. 在选举学生会会长时，一共有 4 名候选人，有 453 名学生可以把自己的一票投给其中一名候选人。已知当选人的票数分别超过其他候选人 31 票、25 票和 19 票。请问当选人一共获得多少张票？

In an election for school captain, there were 4 candidates and 453 students each voted for one candidate. The winner's margins over the other candidates were 31, 25 and 19.
How many votes did the winner receive?

(A) 113　(B) 127　(C) 129　(D) 131　(E) 132

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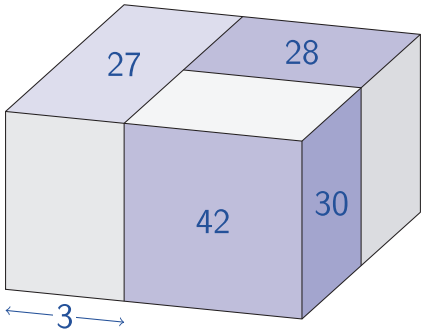
22. 将三个表面都是长方形的木块组合成一个更大的矩形棱柱，如图所示。
已知所有木块的边长均为整数（单位：厘米），图中已标出某些面的面积以及其中一条边的长度。
请问组成的这个矩形棱柱的体积是多少立方厘米？



Three blocks with rectangular faces are placed together to form a larger rectangular prism.
All blocks have side lengths which are whole numbers of centimetres. The areas of some of the faces are shown, as is the length of one edge.
In cubic centimetres, what is the volume of the combined prism?

(A) 360 (B) 540 (C) 600 (D) 720 (E) 900

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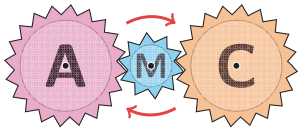
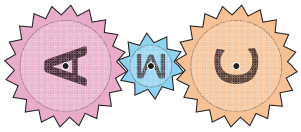
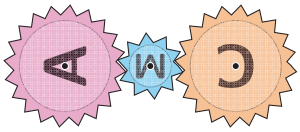
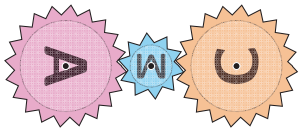
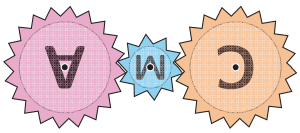
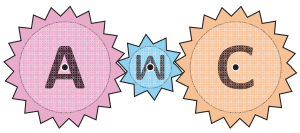
23. 将三个齿轮按照如图所示的方式连接。其中两个较大的齿轮各有 20 个齿、较小的齿轮有 10 个齿。
现将中间的齿轮沿箭头方向旋转半圈，也就是将字母 M 颠倒放置。
经过旋转后，请问三个齿轮看起来会是下列哪种样式？



Three gears are connected as shown. The two larger gears have 20 teeth each and the smaller gear has 10 teeth.
The middle gear is rotated half a turn in the direction of the arrows, turning the M upside down.
What do the three gears look like after this rotation?

(A)

(B)

(C)

(D)

(E)


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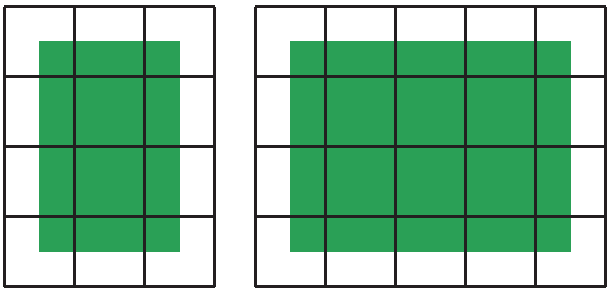
24. 小安有很多以下三种类型的瓷砖。

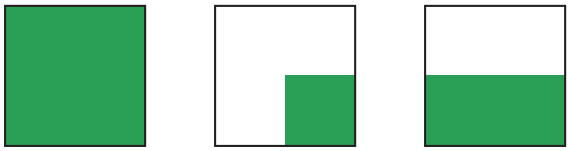
她想拼出一个与下图类似的具有白色边框的绿色矩形瓷砖。



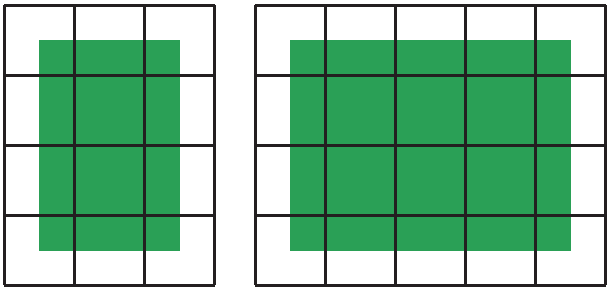
于是她使用尽可能多的瓷砖拼出了一个类似的矩形瓷砖，恰好使用了 20 块全绿的瓷砖。请问她一共使用了多少块瓷砖？

(A) 80 (B) 66 (C) 48 (D) 42 (E) 39

Anna has a large number of tiles of three types:



She wants to build a green rectangle with a white frame similar to those below.



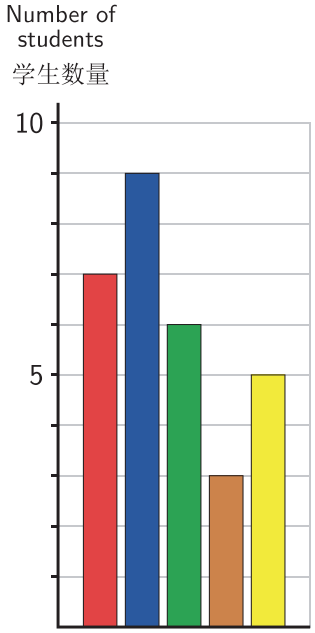
She builds such a rectangle using as many tiles as possible while using exactly 20 completely green tiles. How many tiles will she use altogether?

(A) 80 (B) 66 (C) 48 (D) 42 (E) 39

25. 在小杰的家乡，人们住在北区、东区、南区、西区或中区。小杰正在制作一张表格来统计他班上学生的住处，但不幸的是，在他还没标记好每列的数据时，狗就咬掉了统计数据。

小杰只记得关于数据的两条信息：住在南区的学生比住在东区和中区的学生都多；住在北区和中区的学生总人数和住在其它三个区的学生总人数相同。

仅利用以上信息，请问小杰能正确标记几列数据？



In Jeremy's hometown of Windar, people live in either North, East, South, West or Central Windar.

Jeremy is putting together a chart showing where the students in his class live, but unfortunately his dog chewed his survey results before he managed to label the five columns.

He only remembers two things about the survey: South Windar is more common than both East and Central Windar, and the number of students in North and Central Windar combined is the same as the total of the other three regions.

Using only this information, how many columns can Jeremy correctly label with 100% certainty?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 5

**问题 26-30 的答案为 000-999 之间的整数，
请将答案填在答题卡上对应的位置。**

**第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，
第 29 题占 9 分，第 30 题占 10 分。**

For questions 26 to 30, shade the answer as a whole number from 000 to 999 in the space provided on the answer sheet.

Questions 26 – 30 are worth 6, 7, 8, 9 and 10 marks, respectively.

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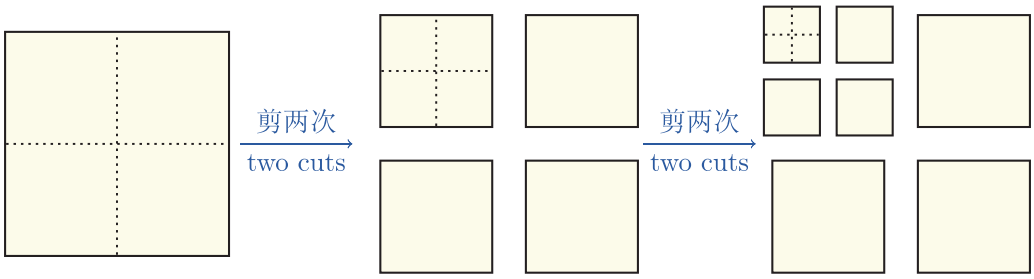
26. 小派沿着两条直线将一张大的正方形纸张裁剪两次，剪成四张较小的正方形。然后取出其中一张小正方形，再剪两次将它剪成四张更小的正方形，如下图所示。



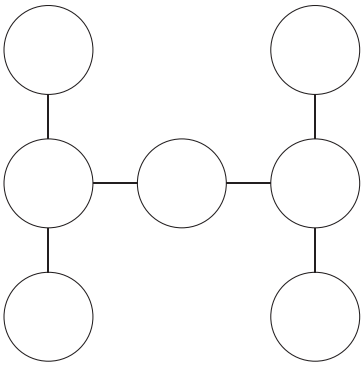
继续重复以上过程，请问小派需要剪多少次才能得到总共 1000 张不同尺寸大小的正方形？

Pip starts with a large square sheet of paper and makes two straight cuts to form four smaller squares. She then takes one of these smaller squares and makes two more straight cuts to make four even smaller ones, as shown.

Continuing in this way, how many cuts does Pip need to make to get a total of 1000 squares of various sizes?

---

27. 从 1 到 9 选出七个数字填入如图所示的圆圈内，使得每条竖线上的数之积和水平线上的数之积相同。请问这一乘积是多少？



Seven of the numbers from 1 to 9 are placed in the circles in the diagram in such a way that the products of the numbers in each vertical or horizontal line are the same.

What is this product?

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28. 一只兔子和一只乌龟参加一场 10 km 的赛跑。兔子的奔跑速度是 30 km/h，乌龟的爬行速度是 3 km/h。但不幸的是，出发时兔子朝着反方向跑去。过了一阵子，它才发现错误开始往回跑，并在路程的一半处追上了乌龟。
请问这只兔子朝错误方向一共跑了多少分钟？

A hare and a tortoise compete in a 10 km race. The hare runs at 30 km/h and the tortoise walks at 3 km/h. Unfortunately, at the start, the hare started running in the opposite direction. After some time, it realised its mistake and turned round, catching the tortoise at the halfway mark.
For how many minutes did the hare run in the wrong direction?

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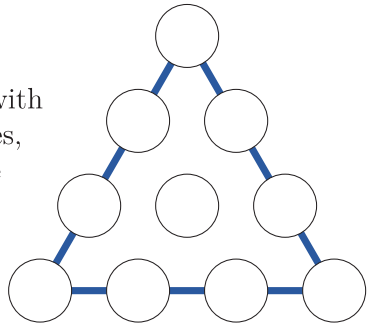
29. 将数 1 到 10 填入右图，其中每个圆圈内的数字都不重复。三条边上的四个数相加之和即为这条边的边长，已知三条边的边长都相同。
请问边长的最小可能值是多少？

I want to place the numbers 1 to 10 in this diagram, with one number in each circle. On each of the three sides, the four numbers add to a side total, and the three side totals are all the same.
What is the smallest number that this side total could be?



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30. 有两个数相加之和为 11.63。在将这两个数相加时，小欧不小心将其中一个数的小数点向左移动了一位，结果小欧计算出的答案为 5.87。
请问原来这两个数之差的一百倍是多少？

The sum of two numbers is 11.63. When adding the numbers together, Oliver accidentally shifted the decimal point in one of the numbers one position to the left. Oliver got an answer of 5.87 instead.
What is one hundred times the difference between the two original numbers?